TECHNICAL SPECIFICATIONS

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ND-526 BIAS POWER SUPPLY

The ND-526 BPS is an ultra-stable bias power supply designed specifically for operation of solid state detectors. It is housed in a standard AEC double width module and operates with standard bin power supply voltages. Detector operating potentials of up to 1400 volts D.C. are obtained by conversion of relatively low D.C. voltage from the bin power supply to 20KHz A.C. voltage by means of an electronic "chopper", the output of which is coupled to a voltage multiplier. The output of the voltage multiplier is shunted by series-connected miniature gaseous reference elements to provide an extremely stable source voltage for constant current loads.

A ten-turn calibrated front panel control provides adjustment of output voltage over a range of 0 to 1400 volts. The maximum rate of change for output voltage adjustment is limited by a 0.5 second time constant. This protects both the detector and the preamplifier input from damage which may result from sudden adjustment-induced voltage changes.

An A.C. RMS voltmeter is also included as a noise indicator for determining optimum detector voltage settings. It can also be used to determine which integrating and differentiating time constants of a linear pulse amplifier, such as the ND-524 LARC, provide optimum signal-to-noise ratio.

BIAS SUPPLY

Output Voltage:
0-1400 volts D.C.

Output Polarity:
Positive (negative output obtained by relocating three internal jumpers).

Output Impedance:
100 K ohms.

Output Noise:
0.4 millivolts rms, from 2Hz to 4MHz.

Bias Stability:
Temperature - 0.007%/°C.
A.C. Line Voltage - 0.0005%/10% change in line voltage.

Bias Control:
Ten turn potentiometer calibrated at 140 volts per turn.

Output Time Constant:
A 0.5 second time constant limits the maximum rate of change of output voltage.

RMS NOISE INDICATOR

Input Impedance:
1,000 ohms (D.C. voltage not to exceed ±15 volts).

Indication:
Front panel meter.

Accuracy:
10%

Sensitivity:
10 mv rms full scale, 100 mv rms full scale, and 1 volt rms full scale, by means of a three position switch.

Band Width:
20 Hz to 2MHz.

GENERAL

Connectors:
Bias Output - UG 931/U (MHV).
RMS Input - UG 290/U (BNC).

Power Requirements:
+12 volts at 150 ma.
−12 volts at 150 ma.
−24 volts at 5 ma.

Size:
8.71" high x 2.68" wide x 9.7" deep (Standard AEC Double Width).

Weight:
4 pounds.

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